

Linux Containers Lab Guide with Command Explanations and Screenshot Placeholders

Step 1 - Pull Ubuntu Image

Command: `docker pull ubuntu:24.04`

Explanation: Downloads the Ubuntu Linux image from Docker Hub to your local machine. Images are templates used to create containers.

Expected Output: `24.04: Pulling from library/ubuntu`

`532a4cf7e47b: Download complete`

Digest:

`sha256:786a8b558f7be160c6c8c4a54f9a57274f3b4fb1491cf65146521ae77ff1dc`

`54`

Status: Downloaded newer image for `ubuntu:24.04`

`docker.io/library/ubuntu:24.04`

Screenshot Placeholder (Paste Screenshot Below):

```
Last login: Fri Jun 19 11:40:20 on tty000
/dev/fd/12:25: command not found: compdef
vaibhavgupta@FVF0985WQ85F~vaibhavgupta Day 41 June 22 % docker pull ubuntu:24.04

docker images

24.04: Pulling from library/ubuntu
532a4cf7e47b: Download complete
Digest: sha256:786a8b558f7be160c6c8c4a54f9a57274f3b4fb1491cf65146521ae77ff1dc54
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu:24.04

IMAGE          ID              DISK USAGE  CONTENT SIZE  EXTRA
alpine:latest  28bd5fe8b66d    13.6MB      4.27MB
compose-postgres-api-lab-api:latest  aebc9500f8bf    387MB      97MB      U
customer-api:1.0  62593972ba26    350MB      88.3MB      U
dpkg/pgadmin4:latest  cefc4cc6b769    765MB      175MB      U
node:22-alpine  65b326a86441    229MB      58.3MB
postgres:16    081f1bc7bd5e    663MB      165MB      U
postgres:latest  29ee7bb3bd88    671MB      167MB
ubuntu:24.04    786a8b558f7b    141MB      38.8MB
weather-viz:1.0  3992a65a2f558    76.1MB      21.2MB
weatherapi:1.0  aefb12c8db89    351MB      88.4MB
weatherapi:single  8cf8eb18e30e    1.25GB     311MB
```

Step 2 - Verify Image

Command: docker images

Explanation: Displays all images stored locally on your computer.

Expected Output: vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 41 June 22

% docker images

i Info → U In Use

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	28bd5fe8b56d	13.6MB	4.27MB	U
compose-postgres-api-lab-api:latest	aebc9500f8bf	387MB	97MB	U
customer-api:1.0	62593972ba26	350MB	88.3MB	U
dpage/pgadmin4:latest	cefc4cc6b7d9	765MB	175MB	U
node:22-alpine	e58326d0d441	229MB	58.5MB	
postgres:16	081f1bc7bd5e	663MB	165MB	U
postgres:latest	29ee7bb30d80	671MB	167MB	U
ubuntu:24.04	786a8b558f7b	141MB	30.8MB	U
weather-ui:1.0	399165a2f358	76.1MB	21.9MB	
weatherapi:1.0	aefb12c8db89	351MB	88.4MB	
weatherapi:single	8cf0eb18e30e	1.25GB	311MB	

Screenshot Placeholder (Paste Screenshot Below):



Vaibhav Gupta (vgupta1@presidio.com)

Step 3 - Start Ubuntu Container

Command: `docker run -it ubuntu:24.04 bash`

Explanation: Creates and starts a container from the Ubuntu image and opens an interactive Bash shell.

Expected Output: `vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 41 June 22`

`% docker run -it ubuntu:24.04 bash`

`root@5f7ddcef7de8:/#`

Screenshot Placeholder (Paste Screenshot Below):

```
[vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 41 June 22 % docker run -it ubuntu:24.04 bash
root@5f7ddcef7de8:/#
```

Step 4 - Check Current Directory

Command: `pwd`

Explanation: Prints the current working directory.

Expected Output: `root@5f7ddcef7de8:/# pwd`

`/`

Screenshot Placeholder (Paste Screenshot Below):

```
[vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 41 June 22 % docker run -it ubuntu:24.04 bash
root@5f7ddcef7de8:/# pwd
/
root@5f7ddcef7de8:/#
```

Step 5 - List Files

Command: `ls -la`

Explanation: Displays all files and folders including hidden files.

Expected Output: `root@5f7ddcef7de8:/# ls -la`

`total 56`

`drwxr-xr-x 1 root root 4096 Jun 22 04:56 .`

`drwxr-xr-x 1 root root 4096 Jun 22 04:56 ..`

Vaibhav Gupta (vgupta1@presidio.com)

```
-rwxr-xr-x  1 root root   0 Jun 22 04:56 .dockerenv
lrwxrwxrwx  1 root root   7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x  2 root root 4096 Apr 22  2024 boot
drwxr-xr-x  5 root root 360 Jun 22 04:56 dev
drwxr-xr-x  1 root root 4096 Jun 22 04:56 etc
drwxr-xr-x  3 root root 4096 May  9 14:23 home
lrwxrwxrwx  1 root root   7 Apr 22  2024 lib -> usr/lib
drwxr-xr-x  2 root root 4096 May  9 14:17 media
drwxr-xr-x  2 root root 4096 May  9 14:17 mnt
drwxr-xr-x  2 root root 4096 May  9 14:17 opt
dr-xr-xr-x 218 root root   0 Jun 22 04:56 proc
drwx----- 2 root root 4096 May  9 14:23 root
drwxr-xr-x  4 root root 4096 May  9 14:23 run
lrwxrwxrwx  1 root root   8 Apr 22  2024/sbin -> usr/sbin
drwxr-xr-x  2 root root 4096 May  9 14:17 srv
dr-xr-xr-x 11 root root   0 Jun 22 04:25 sys
drwxrwxrwt  2 root root 4096 May  9 14:23 tmp
drwxr-xr-x 11 root root 4096 May  9 14:17 usr
drwxr-xr-x 11 root root 4096 May  9 14:23 var
```

Screenshot Placeholder (Paste Screenshot Below):

```
root@5f7ddcef7de8:/# ls -la
total 56
drwxr-xr-x 1 root root 4096 Jun 22 04:56 .
drwxr-xr-x 1 root root 4096 Jun 22 04:56 ..
-rwxr-xr-x 1 root root 0 Jun 22 04:56 .dockerenv
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Apr 22 2024 boot
drwxr-xr-x 5 root root 368 Jun 22 04:56 dev
drwxr-xr-x 1 root root 4096 Jun 22 04:56 etc
drwxr-xr-x 3 root root 4096 May 9 14:23 home
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
drwxr-xr-x 2 root root 4096 May 9 14:17 media
drwxr-xr-x 2 root root 4096 May 9 14:17 mnt
drwxr-xr-x 2 root root 4096 May 9 14:17 opt
dr-xr-xr-x 218 root root 0 Jun 22 04:56 proc
drwx----- 2 root root 4096 May 9 14:23 root
drwxr-xr-x 4 root root 4096 May 9 14:23 run
lrwxrwxrwx 1 root root 8 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 May 9 14:17 srv
dr-xr-xr-x 11 root root 0 Jun 22 04:25 sys
drwxrwxrwt 2 root root 4096 May 9 14:23 tmp
drwxr-xr-x 11 root root 4096 May 9 14:17 usr
drwxr-xr-x 11 root root 4096 May 9 14:23 var
root@5f7ddcef7de8:/#
```

Step 6 - View Linux Version

Command: `cat /etc/os-release`

Explanation: Displays operating system information.

Expected Output: `PRETTY_NAME="Ubuntu 24.04.4 LTS"`

`NAME="Ubuntu"`

`VERSION_ID="24.04"`

`VERSION="24.04.4 LTS (Noble Numbat)"`

`VERSION_CODENAME=noble`

`ID=ubuntu`

`ID_LIKE=debian`

`HOME_URL="https://www.ubuntu.com/"`

`SUPPORT_URL="https://help.ubuntu.com/"`

`BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"`

`PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"`

`UBUNTU_CODENAME=noble`

`LOGO=ubuntu-logo`

Screenshot Placeholder (Paste Screenshot Below):

Vaibhav Gupta (vgupta1@presidio.com)

```
root@5f7ddcef7de8:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@5f7ddcef7de8:/#
```

Step 7 - Check Current User

Command: whoami

Explanation: Displays the currently logged-in user.

Expected Output `root@5f7ddcef7de8:/# whoami`

`root`

Screenshot Placeholder (Paste Screenshot Below):

```
[root@5f7ddcef7de8:/# whoami
root
root@5f7ddcef7de8:/#
```

Step 8 - Create File

Command: touch notes.txt

Explanation: Creates an empty file named notes.txt.

Expected Output:

Screenshot Placeholder (Paste Screenshot Below):

```
bin boot dev etc home lib media mnt
[root@5f7ddcef7de8:/# touch notes.txt
root@5f7ddcef7de8:/#
```

Step 9 - Write Data

Command: echo "Hello Docker" > notes.txt

Explanation: Writes text into a file.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):

```
[root@5f7ddcef7de8:/# touch notes.txt
[root@5f7ddcef7de8:/# echo "Hello Docker" > notes.txt
root@5f7ddcef7de8:/#
```

Step

10 - Create Folder

Command: mkdir training

Explanation: Creates a new directory.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):

```
[root@5f7ddcef7de8:/# echo "Hello Docker" > notes.txt
[root@5f7ddcef7de8:/# mkdir training
root@5f7ddcef7de8:/#
```

Step 11 - View Processes

Command: ps -ef

Explanation: Displays running processes inside the container.

Expected Output:

UID	PID	PPID	C	STIME	TTY	TIME	CMD
-----	-----	------	---	-------	-----	------	-----

root	1	0	0	04:56	pts/0	00:00:00	bash
root	15	1	0	05:01	pts/0	00:00:00	ps -ef

Screenshot Placeholder (Paste Screenshot Below):

```
[root@5f7ddcef7de8:/# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1      0  0 04:56 pts/0    00:00:00 bash
root          15      1  0 05:01 pts/0    00:00:00 ps -ef
root@5f7ddcef7de8:/#
```

Step 12 - Install Software

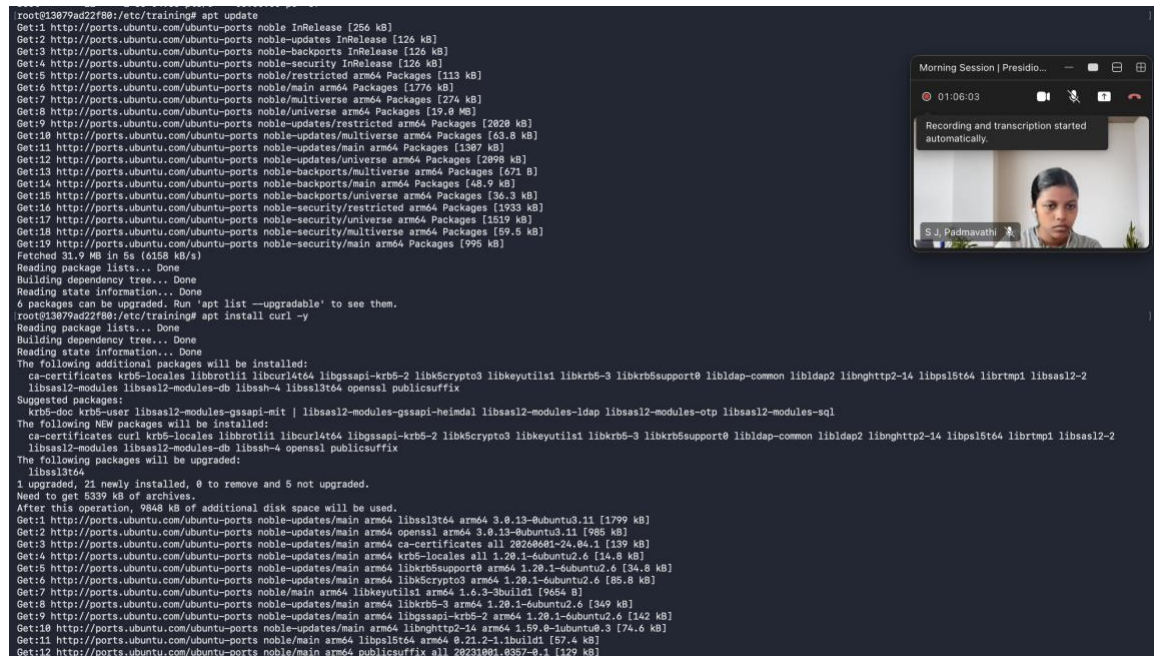
Commands: apt update && apt install curl -y

Vaibhav Gupta (vgupta1@presidio.com)

Explanation: Updates package repositories and installs curl.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):



Step 13 - Networking

Commands: hostname, hostname -I

Explanation: Displays hostname and IP address.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):

Step 14 - Create User

Command: useradd student

Vaibhav Gupta (vgupta1@presidio.com)

Explanation: Creates a new Linux user.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):

```
root      15      1  0 05:01 pts/0    00:0
[root@5f7ddcef7de8:/# useradd student
root@5f7ddcef7de8:/#
```

Step 15 - Permissions

Command: `chmod 777 secret.txt`

Explanation: Changes file permissions to full access.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):

```
[root@5f7ddcef7de8:/# chmod 777 secret.txt
chmod: cannot access 'secret.txt': No such file or directory
[root@5f7ddcef7de8:/# chmod 777 notes.txt
[root@5f7ddcef7de8:/# ls -ls
total 60
0 lrwxrwxrwx   1 root root    7 Apr 22  2024 bin -> usr/bin
4 drwxr-xr-x   2 root root 4096 Apr 22  2024 boot
0 drwxr-xr-x   5 root root  360 Jun 22  04:56 dev
8 drwxr-xr-x   1 root root 4096 Jun 22  05:03 etc
4 drwxr-xr-x   3 root root 4096 May  9  14:23 home
0 lrwxrwxrwx   1 root root    7 Apr 22  2024 lib -> usr/lib
4 drwxr-xr-x   2 root root 4096 May  9  14:17 media
4 drwxr-xr-x   2 root root 4096 May  9  14:17 mnt
4 -rwxrwxrwx   1 root root   13 Jun 22  05:01 notes.txt
```

Step 16 - Environment Variables

Command: `export COMPANY=Training`

Explanation: Creates an environment variable.

Expected Output: _____

Screenshot Placeholder (Paste Screenshot Below):

Vaibhav Gupta (vgupta1@presidio.com)

```
[root@5f7ddcef7de8:/# export COMPANY=Training  
root@5f7ddcef7de8:/# █
```

Step 17 - Exit Container

Command: exit

Explanation: Leaves the container shell.

Expected Output: exit

Screenshot Placeholder (Paste Screenshot Below):

```
[root@5f7ddcef7de8:/# export COMPANY=Training  
[root@5f7ddcef7de8:/# exit  
exit  
vaibhavgupta@FVFG905WQ05F-vaibhavgupta Day 41 June 22 % █
```